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## Question 1

Not answered

Marked out of 3.00

**Study designs I**

For each of the descriptions below, select the most appropriate study design from the list provided

To investigate the relationship between domestic abuse and PTSD, patients referred to a psychiatric outpatient clinic with diagnosed PTSD were enquired about domestic abuse. A group of age and sex matched diabetic patients from a medical clinic were also questioned about domestic abuse in the same manner.

A carefully selected group of parents of children attending Child and Adolescent clinic were invited to provide their views on waiting time for psychotherapy and choices of treatments offered.

A survey was conducted in an unsystematic sample of ward staff at a Mental Health Trust. This survey set out to find the degree of satisfaction with jobs at inpatient psychiatric units

Your answer is incorrect.

Explanation:

Q1. In this study, patients with PTSD - the cases - are recruited alongside those with no PTSD - the diabetic patients who act as controls. Hence, this is a case-control study.

Q2. A purposive sample is recruited when opinions/themes are sought in a qualitative study.

Q3. Surveys are almost always cross-sectional.

The correct answer is: To investigate the relationship between domestic abuse and PTSD, patients referred to a psychiatric outpatient clinic with diagnosed PTSD were enquired about domestic abuse. A group of age and sex matched diabetic patients from a medical clinic were also questioned about domestic abuse in the same manner. → Case-control study, A carefully selected group of parents of children attending Child and Adolescent clinic were invited to provide their views on waiting time for psychotherapy and choices of treatments offered. → Qualitative study, A survey was conducted in an unsystematic sample of ward staff at a Mental Health Trust. This survey set out to find the degree of satisfaction with jobs at inpatient psychiatric units → Cross-sectional Study

## Question 2

Not answered

Marked out of 3.00

**Types of Bias I**

For each of the questions below, chose one bias from the list provided:

Intention to treat analysis aims to eliminate this bias

Blinding reduces this bias

Cohort studies are especially prone to this bias

Your answer is incorrect.

Explanation:

Q1. ITT analysis aims to reduce the impact of dropouts and attrition in prospective trials.

Q2. The aim of blinding is to reduce measurement bias.

Q3. Cohort studies are very much prone to attrition bias as long duration of follow-up means a higher risk of 'loss to follow-up' i.e. attrition.

The correct answer is: Intention to treat analysis aims to eliminate this bias → Attrition bias, Blinding reduces this bias → Measurement bias, Cohort studies are especially prone to this bias → Attrition bias

## Question 3

Not answered

Marked out of 4.00

**Qualitative research**

A qualitative research study is conducted to obtain the views of patients with schizophrenia regarding the nature of mental health care they received. From the list of options above choose the most appropriate method for proceeding with this study for each question below:

Method used to reach adequate sample size

You include an independent rater to look at the data collected when analysing it.

You ask psychologists and social workers who provide care to the participating patients to be the study subjects to achieve a wider perspective

You want a systematic approach to analyse the data as and when it is collected.

Your answer is incorrect.

Explanation:

Q1. To reach adequate sample size and stop collecting data, the point of theoretical saturation is employed in qualitative research. The data is iteratively analysed and at the point where no more new categories emerge, the data collection ceases. This point is called as theoretical saturation.

Q2. Triangulation compares the results from either two or more different methods of data collection or two or more data sources. The researcher looks for patterns of convergence to develop or corroborate an overall interpretation. In investigator triangulation, more than one investigator is involved, and findings are cross-validated.

Q3. Purposive sampling is one of the most common sampling strategies in qualitative research. (Note that snowball sampling and quota sampling are also used often). In purposive sampling the participants are selected according to a preselected criterion relevant to a particular research question - hence it is not random. "Sample sizes, which may or may not be fixed prior to data collection, depend on the resources and time available, as well as the study's objectives. Purposive sample sizes are often determined on the basis of theoretical saturation (the point in data collection when new data no longer bring additional insights to the research questions). Purposive sampling is, therefore, most successful when data review and analysis are done in conjunction with data collection"

(Retrieved from <http://www.ccs.neu.edu/course/is4800sp12>). Q4. Grounded theory refers to a constant comparison of the collected data alongside the data collection (iterative analysis). (Excerpt retrieved from Mays & Pope, BMJ 2000;320;50-52)

The correct answer is: Method used to reach adequate sample size → Theoretical saturation, You include an independent rater to look at the data collected when analysing it. → Triangulation, You ask psychologists and social workers who provide care to the participating patients to be the study subjects to achieve a wider perspective → Purposive sampling, You want a systematic approach to analyse the data as and when it is collected. → Grounded theory



## Question 4

Not answered

Marked out of 4.00

## Meta-analysis of antidepressants

Geddes, J. R., Carney, S. M., Davies, C., et al. (2003) Relapse prevention with antidepressant drug treatment in depressive disorders: a systematic review. The Lancet, 361, 653-661

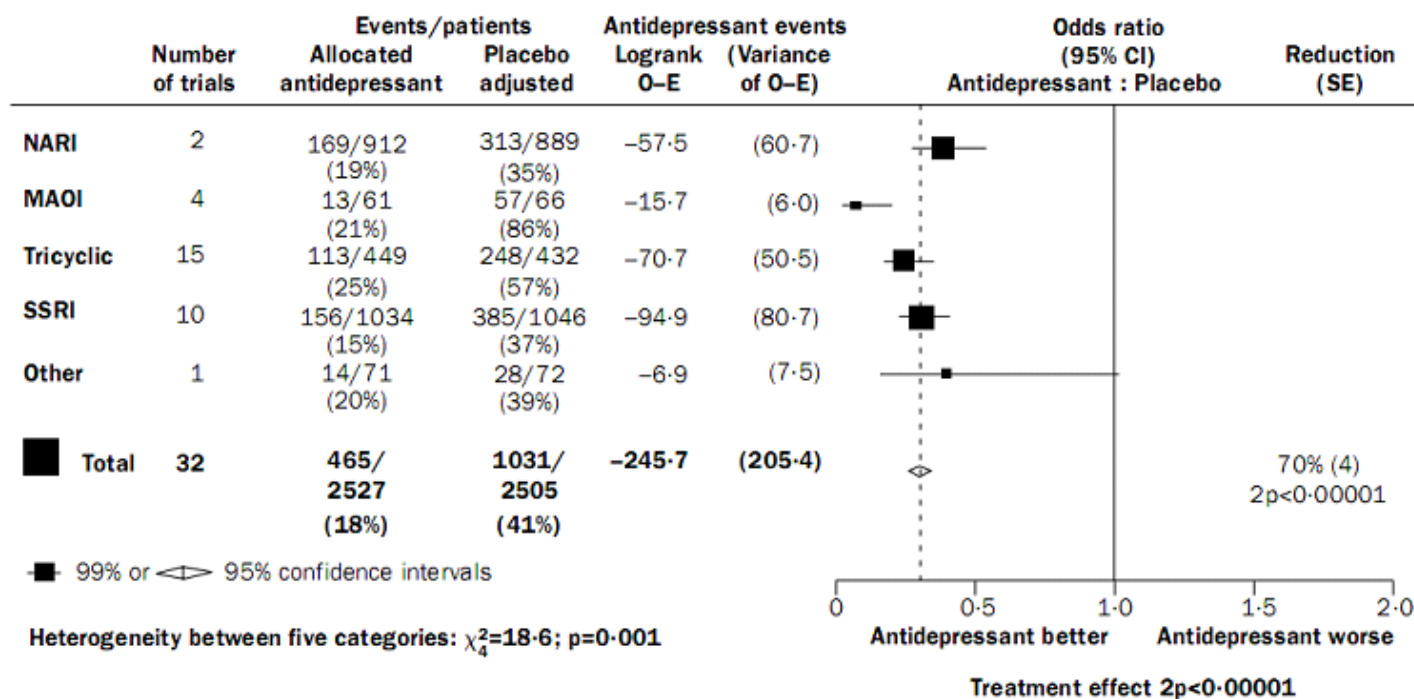


Figure 1: Risk of relapse: odds ratios by drug type

NARI=noradrenaline (norepinephrine) reuptake inhibitor. MAOI=monoamine oxidase inhibitor. SSRI=selective serotonin reuptake inhibitor.

To establish how long antidepressants should be continued to prevent relapse, Geddes et al. (2003) undertook a systematic overview of evidence from 31 randomised trials (4410 participants). The above forest plot is obtained from this study. Please answer the following EMI using this graph.

Choose one correct value from the list below for each question that follows.

The total experimental event rate

Choose...

The control event rate for MAOIs

Choose...

Number needed to treat with any antidepressant to prevent one relapse

Choose...

Relative benefit increase when treated with SSRIs

Choose...

Your answer is incorrect.

Explanation:

Q1. From the above graph, total pooled relapse rate is 18% for antidepressants and 42% for placebo. So, the experimental event rate is 0.18.

Q2. For those who entered RCTs for MAOI as intervention, 86% had a relapse if they did not receive the antidepressants. Hence, CER is 0.86.

Q3. NNT can be calculated from ARR, which is 41%-18% for the pooled data. ARR = 23%. So NNT, which is the reciprocal of ARR, is nearly 4.

Q4. 37% of the placebo group (CER) and 15% SSRI group (EER) relapse in SSRI trials. Hence Relative Risk reduction (or Relative benefit increase) is  $CER - EER / CER = 22/37 = 0.59$

The correct answer is: The total experimental event rate → 0.18, The control event rate for MAOIs → 0.86, Number needed to treat with any antidepressant to prevent one relapse → 4, Relative benefit increase when treated with SSRIs → 0.58

## Question 5

Not answered

Marked out of 4.00

**Study designs II**

Choose the most ethically appropriate and practically possible study design from the list for each research question below

A researcher is interested in studying how many deaths due to suicide are attributable to an overdose of non-prescribed medications in the last three years.

A local PCT asks its paediatric team to look at the number of children with cardiac abnormalities born to mothers on lithium.

A research team working on the neurobiology of psychosis wants to look at obstetric complications and the development of schizophrenia but wants to avoid heavy attrition that is seen in other such studies.

A research team working on the neurobiology of psychosis wants to look at obstetric complications and the development of schizophrenia but wants to avoid recall bias.



Your answer is incorrect.

Explanation:

Q1. The most plausible manner in which this data could be collected is by surveying existing records in a cross-sectional fashion. In fact the National Confidential Inquiry into Suicide and Homicide used this approach, which collected information on people who died by suicide (or who received an open verdict at coroner's inquest) from the Office for National Statistics (ONS) for England and Wales, and the General Register Offices (GRO) in Northern Ireland. This type of inquiry does not involve collecting equivalent information on "controls", individuals who have not committed suicide.

Q2. This requires a 'prevalence' data, which can be obtained by surveying birth records cross-sectionally.

Q3. Cohort studies are very much prone to attrition bias as long duration of follow-up means a higher risk of 'loss to follow-up' i.e. attrition. In this case, we can opt for case control study design.

Q4. Prospective cohort studies circumvent the issue of recall bias.



The correct answer is: A researcher is interested in studying how many deaths due to suicide are attributable to an overdose of non-prescribed medications in the last three years. → Cross-sectional survey, A local PCT asks its paediatric team to look at the number of children with cardiac abnormalities born to mothers on lithium. → Cross-sectional survey, A research team working on the neurobiology of psychosis wants to look at obstetric complications and the development of schizophrenia but wants to avoid heavy attrition that is seen in other such studies. → Case-control study, A research team working on the neurobiology of psychosis wants to look at obstetric complications and the development of schizophrenia but wants to avoid recall bias. → Prospective cohort



## Question 6

Not answered

Marked out of 4.00

**Study designs III**

Choose the most ethically appropriate and practically possible study design from the list for each research question below:

A very costly intervention is introduced as a control measure of multiple sclerosis though it does not cure it. A neurology team wants to compare this treatment with placebo, but they want to recruit as few participants as possible.

A clinical researcher wants to compare whether a new drug has fewer side effects than another established drug.

A nursing research team wishes to get patient's perspectives on the quality of care they receive as inpatients.

A researcher wants to evaluate four treatments for psychotic depression - antidepressants alone, antipsychotics alone, antidepressants and antipsychotics together, neither (placebo).

Your answer is incorrect.

Explanation:

Please remember that most RCTs have a parallel design. In RCTs with parallel group design, each group of participants is exposed to only one of the study interventions. This is useful for comparing both beneficial and adverse effects of either an active drug against placebo (Q2) or two or more active drugs. An RCT has a cross-over design when each of the participants is given all the study interventions in successive periods, with the order being determined at random. Crossover trials produce within participant comparisons, whereas parallel designs produce between participant comparisons. As each participant is also a control in crossover trials, valid results can be obtained with fewer participants than would be required with a parallel design. Hence, this design is likely to be the most suited for Q1. When two or more experimental interventions are not only compared separately but also in combination and against a control in an RCT, then it is called a factorial RCT. For Q3 where 'values' are measured, a qualitative design is appropriate.

The correct answer is: A very costly intervention is introduced as a control measure of multiple sclerosis though it does not cure it. A neurology team wants to compare this treatment with placebo, but they want to recruit as few participants as possible. → Crossover trial, A clinical researcher wants to compare whether a new drug has fewer side effects than another established drug. → Parallel RCT, A nursing

research team wishes to get patient's perspectives on the quality of care they receive as inpatients. → Qualitative study, A researcher wants to evaluate four treatments for psychotic depression - antidepressants alone, antipsychotics alone, antidepressants and antipsychotics together, neither (placebo). → Factorial RCT



## Question 7

Not answered

Marked out of 4.00

**Types of Bias II**

Choose one best option from the list for each description below:

An early intervention team studying the association between risk behaviours such as self-harm and violence in first episode schizophrenia recruits patients from acute admission ward.

In a study exploring the association of bullying at school and adult depression, depressed patients report more bullying than non-depressed controls matched for age and social class.

In a study exploring the development of schizophrenia in mothers exposed to ergot fungus contaminated rye bread only 33 out of 78 children born to the mothers enrolled initially remain in the study after 20 years. Others are lost for follow up.

Due to lack of resources, a research team chooses to use the same research worker to administer screening questionnaire for ADHD (to parents) and carry out video-tape scoring of patients to determine the usefulness of screening questionnaire.

Your answer is incorrect.

Explanation:

In 1946, Joseph Berkson highlighted that the relative frequency of disease in a group of patients who are hospitalised is inherently biased when compared to the population served by the hospital (**Berkson Bias**). This phenomenon is attributable to the way in which the probabilities of hospitalisation combine in patients with more than one disease (if you have two diseases, your probability of being hospitalised is greater than the probability associated with either disease separately). If, for example, obese people who have hypertension have a higher probability of being hospitalised than people with hypertension alone, a spurious association between obesity and anti-hypertensive drugs can be found. People with self-harm or violence are likely to be over-represented in the psychiatric wards, and this over-representation will affect the aim of the study described in Q1.

Q2 describes recall bias, seen in case-control studies. Q3 describes attrition bias seen in longitudinal studies.

A late look at those exposed (or affected) early will miss fatal cases and short episodes, plus mild or silent cases and cases in which evidence of exposure disappears with disease onset. This is called as the **Neyman bias** or the prevalence-incidence bias.

**Diagnostic suspicion bias** is described as the knowledge of the subject's prior exposure to a putative cause influencing both the intensity and the outcome of the diagnostic process. This is likely to occur in scenario 4.

(Retrieved from <http://www.chnri.org/resources/1.%20Learning%20Resource%20Material/Epidemiology/>)

The correct answer is: An early intervention team studying the association between risk behaviours such as self-harm and violence in first episode schizophrenia recruits patients from acute admission ward. → Berkson bias, In a study exploring the association of bullying at school and adult depression, depressed patients report more bullying than non-depressed controls matched for age and social class. → Recall bias, In a study exploring the development of schizophrenia in mothers exposed to ergot fungus contaminated rye bread only 33 out of 78 children born to the mothers enrolled initially remain in the study after 20 years. Others are lost for follow up. → Attrition bias, Due to lack of resources, a research team chooses to use the same research worker to administer screening questionnaire for ADHD (to parents) and carry out video-tape scoring of patients to determine the usefulness of screening questionnaire. → Diagnostic suspicion bias



## Question 8

Not answered

Marked out of 3.00

**Research Methods I**

From the options listed here, select the most appropriate ethical study design to address the following questions.

You want to find out whether a new short-acting analgesic is likely to be more effective at relieving chronic pain without troublesome side-effects. You also want to recruit as few people as possible without losing statistical power.

Following a 6-month placebo-controlled trial of donepezil for Alzheimer's disease, researchers want to know whether mortality is reduced at one year in those continuing to take donepezil.

For people with a newly diagnosed bipolar disorder, is lithium, divalproex sodium, or a combination the most appropriate treatment to reduce the risk of relapse?



Your answer is incorrect.

Explanations: As each participant is also a control in **crossover trials**, valid results can be obtained with fewer participants than would be required with a parallel design. Hence, this design is likely to be the most suited for Q1.

There are three apparent purposes for continuing a study without any blinding after the conclusion of an RCT. The first reason is simply to make the (now known to be) effective but as yet unlicensed drug available to participants who were randomized to placebo; this might have been a requirement of the ethics approval or a means of enhancing recruitment to the original RCT. A second reason is that further, more prolonged observation may disclose adverse effects that were not observed in the original parent RCT. The third purpose may be to demonstrate continued efficacy of the drug over a longer period of time (as in Q2) or to show that participants randomized to receive the active treatment during the open-label phase achieved outcomes similar to those of participants who received the drug from the beginning of the parent RCT.

A simple, parallel RCT is sufficient for the scenario described in Q3.

(What Are Open-Label Extension Studies For? Excerpt retrieved from <http://www.jrheum.com/subscribers/06/04/642.html>)

The correct answer is: You want to find out whether a new short-acting analgesic is likely to be more effective at relieving chronic pain without troublesome side-effects. You also want to recruit as few people as possible without losing statistical power. → Cross-over randomised controlled trial, Following a 6-month placebo-controlled trial of donepezil for Alzheimer's disease, researchers want to know whether mortality is reduced at one year in those continuing to take donepezil. → Open label extension of a Randomised Controlled Trial, For people with a newly diagnosed bipolar disorder, is lithium, divalproex sodium, or a combination the most appropriate treatment to reduce the risk of relapse? → Parallel group Randomised Controlled Trial



## Question 9

Not answered

Marked out of 4.00

**Research Methods II**

From the options listed below, select the most appropriate ethical study design to address the following questions.

You want to find out the most appropriate cut-point for the Hamilton Depression Rating Scale to screen for depression. You want to use DSM-IV diagnosis as the reference standard.

You are not convinced that teaching general practitioners the art of motivational interviewing improves abstinence among harmful drinkers in primary care. How can this be tested?

You want to determine whether there is an association between the gross domestic product of different countries and the prevalence of schizophrenia

You want to find out whether cognitive behavioural therapy is helpful in treating people with peanut butter phobia. Only five people are identified with this condition express a wish to take part in the proposed research.

Your answer is incorrect.

Explanations:

Screening tools are often evaluated using cross-sectional designs (Q1).

Cluster RCTs are particularly useful in general practice where the cluster is the general practitioner or the practice (Q2). For example, in the Diabetes Care from Diagnosis Trial, general practitioners were randomised to be trained in patient-centered care or not. All patients under the care of one general practitioner receive the same treatment and so cannot be considered to be independent items. One of the main reasons for conducting cluster trials is fear of contamination, whereby patients used as controls are exposed to the intervention. In ecological studies, the unit of observation is the population or community.

Common types of ecological study are geographical comparisons, time trend analysis or studies of migration (Q3).



A case series is a descriptive study of a group of people, who usually receive the same treatment or who have the same disease. This type of study can describe characteristics or outcomes in a particular group of people, but cannot determine how they compare with people who are treated differently or who do not have the condition (Q4). (Excerpt from BMJ 2004; 328:702-708)

The correct answer is: You want to find out the most appropriate cut-point for the Hamilton Depression Rating Scale to screen for depression. You want to use DSM-IV diagnosis as the reference standard. → Cross-sectional study, You are not convinced that teaching general practitioners the art of motivational interviewing improves abstinence among harmful drinkers in primary care. How can this be tested? → Cluster randomised trial, You want to determine whether there is an association between the gross domestic product of different countries and the prevalence of schizophrenia → Ecological study, You want to find out whether cognitive behavioural therapy is helpful in treating people with peanut butter phobia. Only five people are identified with this condition express a wish to take part in the proposed research. → Case series



## Question 10

Not answered

Marked out of 4.00

**Types of RCT (1)**

Identify the type of RCT based on the descriptions given below

Each participant is given both the experimental and the standard intervention in succession

Choose...

2 or more experimental interventions are evaluated separately and in combination.

Choose...

Each group is exposed to only one of the interventions.

Choose...

There is only one participant in this design

Choose...

Your answer is incorrect.

Explanation:

In parallel design RCT, each group is exposed to only one of the interventions.

In crossover design RCT, each participant is given all the interventions in succession. The order in which the interventions are given is generated randomly.

In factorial design RCT, 2 or more experimental interventions are evaluated separately and in combination. e.g. Psychotherapy vs. Medication vs. Psychotherapy plus Medication vs. neither psychotherapy nor medication.

In n of 1 RCT (individual patient trials), there is only one participant in the trial.

The correct answer is: Each participant is given both the experimental and the standard intervention in succession → Crossover design RCT, 2 or more experimental interventions are evaluated separately and in combination. → Factorial design RCT, Each group is exposed to only one of the interventions. → Parallel design RCT, There is only one participant in this design → n of 1 RCT

## Question 11

Not answered

Marked out of 4.00

**Types of RCT (2)**

Identify the type of RCT based on the definitions given below

The participant and investigator doing the outcome measures are blind.

Choose...

The participant is unaware of the intervention he is receiving.

Choose...

The participant and the investigator(s) analyzing the results are also blind

Choose...

Participants are recruited into each group until the investigators can clearly determine the difference or lack of difference between the groups.

Choose...

Your answer is incorrect.

Explanation: In **single-blind RCT** usually the participant is unaware of the intervention he/she is receiving. In **double-blind RCT**, the investigator(s) measuring the outcome is also blind. In **triple-blind RCT** the investigator(s) statistically analyzing the results is also blind. In **sequential RCT**, the participants are recruited into each group until the investigators can clearly determine the difference or lack of difference between the groups. The sample size is not fixed a priori, and the analysis is regularly done, and the trial is stopped by a stopping rule, leading to a lower cost associated with the trial.

The correct answer is: The participant and investigator doing the outcome measures are blind. → Double-blind RCT, The participant is unaware of the intervention he is receiving. → Single-blind RCT, The participant and the investigator(s) analyzing the results are also blind → Triple-blind RCT, Participants are recruited into each group until the investigators can clearly determine the difference or lack of difference between the groups. → Sequential RCT

## Question 12

Not answered

Marked out of 3.00

**Intention to treat**

Identify the methods of intention to treat in an RCT from the descriptions given below

Assume drop-outs in active arm have negative outcomes and drop-outs in control arm have positive outcomes

Choose...

Assume there was no change in the available ratings after drop-out

Choose...

Assume that the scores for the drop-out subjects were same as the average for their groups

Choose...

Your answer is incorrect.

Explanation: The methods of intention to treat include 1) **worst-case scenario**: Assume drop-outs in active arm have negative outcomes and dropouts in control arm have positive ones 2) **Last observation carried forward approach** assumes that the last score in a rating scale available from a subject who dropped out continued till the end of the trial and had no further change. 3) **Mean imputations**- Assume that the score for a subject who dropped out is same as the average score in a rating scale available from the rest of the group (Ref: Oxford handbook of psychiatry- pg 110)

The correct answer is: Assume drop-outs in active arm have negative outcomes and drop-outs in control arm have positive outcomes → Worst case scenario, Assume there was no change in the available ratings after drop-out → Last observation carried forward, Assume that the scores for the drop-out subjects were same as the average for their groups → Mean imputations

## Question 13

Not answered

Marked out of 4.00

**Sampling techniques for qualitative research**

A researcher wants to explore service users' perceptions and attitudes associated with the prescription and administration of clozapine in community mental health settings. He chooses to use one of the following sampling methods. Choose the correct term describing each approach from the list below.

The researcher chooses to offer a semi-structured interview to the patients who attend clozapine monitoring clinic once every three months. All active community clozapine users are registered to attend this clinic.

The researcher pursues a set of service users who experienced an unusual increase in sweating with clozapine. Another service user who shared the same effect introduced these patients to the researcher.

The researcher specifically attempts to include some highly satisfied service users, some poorly compliant users, and some users who discontinued early in the course and some users who continue to have psychotic symptoms despite high degree of concordance.

The researcher recruits four service-user experts who have wide experience of talking to clozapine users in the locality.

Your answer is incorrect.

**Convenience (or opportunistic) sampling** is a technique that uses an open period of recruitment that continues until a set number of subjects, events, or institutions are enrolled. Here, the selection is based on a first-come, first-served basis. This approach is used in studies drawing on predefined populations such as participants in support groups or medical clinics. Second, **purposive sampling** is a practice where subjects are intentionally selected to represent some explicit predefined traits or conditions. This is analogous to stratified samples in probability-based approaches. The goal here is to provide for relatively equal numbers of different elements or people to enable exploration and description of the conditions and meanings occurring within each of the study conditions. The objective, however, is not to determine the prevalence, incidence, or causes. Third, **snowballing** or word-of-mouth techniques make use of participants as referral sources. Participants recommend others they know who may be eligible. Fourth, case study samples select a single individual, institution, or event as

the total sample. A variant is the key-informant approach or **intensity sampling** (Patton 1990) where a subject who is an expert in the topic of study serves to provide expert information on the specialized topic. When qualitative perspectives are sought as part of clinical or survey studies, the purposive or case study sampling techniques are generally the most useful.

The correct answer is: The researcher chooses to offer a semi-structured interview to the patients who attend clozapine monitoring clinic once every three months. All active community clozapine users are registered to attend this clinic. → Concordant sampling, The researcher pursues a set of service users who experienced an unusual increase in sweating with clozapine. Another service user who shared the same effect introduced these patients to the researcher. → Snowballing, The researcher specifically attempts to include some highly satisfied service users, some poorly compliant users, and some users who discontinued early in the course and some users who continue to have psychotic symptoms despite high degree of concordance. → Purposive sampling, The researcher recruits four service-user experts who have wide experience of talking to clozapine users in the locality. → Intensity sampling



Question **14**

Not answered

Marked out of 3.00

**Epidemiological concepts 1**

Identify the correct term for each of the following definitions:

The proportion of a population who have the disease of interest at a given point in time.

The proportion of the population that has a history of the disease at some point in their life.

The prevalence during a specified period.

Your answer is incorrect.

Explanation:

- 1) Point prevalence is the proportion of a defined population known to have a disease at a particular point in time
- 2) Lifetime prevalence is the proportion of the population that has a history of the disease at some point in their life.
- 3) Period prevalence is the prevalence during a specified period of time.

The correct answer is: The proportion of a population who have the disease of interest at a given point in time. → Point prevalence, The proportion of the population that has a history of the disease at some point in their life. → Lifetime prevalence, The prevalence during a specified period. → Period prevalence



## Question 15

Not answered

Marked out of 3.00

**Epidemiological concepts 2**

Identify the correct term for each of the following definitions:

The number of new cases in a population during a specified period.

Choose...

The complete set of subjects from which a sample is drawn

Choose...

The source of subjects available for inclusion in a sample for an epidemiological study

Choose...

Your answer is incorrect.

Explanation: **Incidence** is the number of new cases in a population during a specified period of time. It is also known as the incidence rate. The term **population** refers to the complete set of subjects from which a sample is drawn. The **sampling frame** refers to the set of elements from which a researcher can select a sample of the intended target population. A researcher does not usually have a direct access to the entire population but tends to rely on the sampling frame to gather a representative sample. (Ref: Making sense of critical appraisal- 203)

The correct answer is: The number of new cases in a population during a specified period. → Incidence, The complete set of subjects from which a sample is drawn → Population, The source of subjects available for inclusion in a sample for an epidemiological study → Sampling frame



## Question 16

Not answered

Marked out of 3.00

**Measurement scales**

For each of the following scenarios choose the correct option from the responses below

You want to measure objective improvement in the mental state of a group of depressed individuals two weeks after an intervention

You want to assess change in the subjective severity of depression four weeks after an intervention in a person with moderate intellectual disability

You want to develop a single index that reflects people's willingness to have socially close contact with immigrants

Your answer is incorrect.

Explanation:

Scales such as CGI consist of items that are based on **Likert approach**, with a verbal description attached to each point of the scale. These are useful in the objective measurement of change in the symptom status after an intervention.

A **Guttman Scalogram** is a scale composed of several yes/no items in a manner such that if a respondent agrees to the one statement, he/she is likely to agree to the lower order statements as well. So the total score of the scale represents a single index for the continuum that is measured. In other words, a person's entire set of responses to all items can be reconstructed from their score. This procedure is also called cumulative scaling. The Guttman scale is most suitable for hierarchical constructs e.g. social attitudes, political views, developmental stages, etc. E.g. Bogardus Social Distance Scale is a cumulative scale used to measure the willingness of people in accepting an immigrant into their social groups (E.g., family, work etc).

**Visual analogue scales** or pictorial thermometers are often used to assess self-report measures for pain, depression and other mental states in adults and children. These scales follow a Likert-type pattern but use either numerical or intensity coding for grading a continuum. For children below the age of 6, and adults with learning disabilities, number/word free versions with smiley faces can be used to generate self-report measures.

The correct answer is: You want to measure objective improvement in the mental state of a group of depressed individuals two weeks after an intervention → Verbally descriptive Likert scale, You want to assess change in the subjective severity of depression four weeks after an

intervention in a person with moderate intellectual disability → Visual analogue scale, You want to develop a single index that reflects people's willingness to have socially close contact with immigrants → Guttman scalogram



## Question 17

Not answered

Marked out of 4.00

**Investigations in schizophrenia**

Choosing investigational approaches in schizophrenia epidemiology

Separating environmental factors from genetic factors

Choose...

The presence of neurological or motor abnormalities that predate onset of illness

Choose...

Identification of cognitive and IQ deficits before the illness onset

Choose...

Delineating the natural course of the illness

Choose...

Your answer is incorrect.

Explanation:

The observation that schizophrenia is more commonly observed among the relatives of individuals with schizophrenia than in the general population could be attributed to both genetic and shared environmental causes. **Adoption designs** permit evaluation of the role of genetic factors in schizophrenia independently of the influence of family environments.

Walker and Levine (1998) successfully employed the analysis of childhood **home movies** by blind raters to identify children who later developed schizophrenia on the basis of motor abnormalities. This study suggested that preschizophrenic subjects can be distinguished from sibling control subjects within the first eight years of life by observing their behavior. Am J Psychiatry 1990;147:1052-1056.

To examine associations between premorbid cognitive functioning in childhood or adolescence and subsequent schizophrenia, three broad categories of study have been employed: birth cohort designs with cognitive testing during childhood, **army conscript designs** with cognitive performance measured at conscription, and studies using school grades. (McCabe, Epidemiol Rev (2008) 30 (1): 77-83.)

To study the natural course of illness, without any alterations induced by pharmacological or treatment agents, **unmedicated cohorts** must be observed prospectively.

The correct answer is: Separating environmental factors from genetic factors → Adoption design, The presence of neurological or motor abnormalities that predate onset of illness → Home movies, Identification of cognitive and IQ deficits before the illness onset → Study of

army conscripts, Delineating the natural course of the illness → Unmedicated cohorts



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